Linked List Practice Questions (10 Questions)

Question 1:
Create a singly linked list and display all its elements.
Examples:
Input: `10 → 20 → 30 → NULL`
Output: `10 20 30`

Question 2:
Count the total number of nodes in a singly linked list.
Examples:
Input: `5 → 8 → 12 → NULL`
Output: `3`

Question 3:
Display the elements of a linked list in reverse order using recursion (without modifying the list).
Examples:
Input: `10 → 20 → 30 → NULL`
Output: `30 20 10`

Question 4:
Search for a given element in a linked list and print its position if found.
Examples:
Input: `List: 10 → 25 → 30 → 45 → NULL`, Search element: `30`
Output: `Element 30 found at position 3`

Question 5:
Find the sum of all node values in a singly linked list.
Examples:
Input: `2 → 4 → 6 → 8 → NULL`
Output: `20`

Question 6:
Insert a new node at the beginning of a singly linked list.

Examples:
Input: `List: 20 → 30 → 40 → NULL`, Insert: `10`
Output: `10 → 20 → 30 → 40 → NULL`

Question 7:
Delete a node from the end of a singly linked list.
Examples:
Input: `List: 10 → 20 → 30 → NULL`
Output: `10 → 20 → NULL`

Question 8:
Delete a node at a specific position (given by the user).
Examples:
Input: `List: 5 → 10 → 15 → 20 → NULL`, Position: `3`
Output: `5 → 10 → 20 → NULL`

Question 9:
Reverse a linked list using the iterative method.
Examples:
Input: `1 → 2 → 3 → 4 → NULL`
Output: `4 → 3 → 2 → 1 → NULL`

Question 10:
Merge two sorted linked lists into one sorted linked list.
Examples:
Input:
List 1: `1 → 3 → 5 → NULL`
List 2: `2 → 4 → 6 → NULL`
Output: `1 → 2 → 3 → 4 → 5 → 6 → NULL`
